COM-480 – Milestone 1

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Dataset

Our chosen dataset is the RAWG [3] video games dataset, which contains 54 columns of data dumped from the RAWG API compiling the largest open video game database for a total of 889793 video games spanning 1962-2025 release dates on 50+ platforms.

Each game has basic metrics such as release date, genres, platforms, tags, developers, publishers, ESRB rating, RAWG user rating (0–5) with full distribution breakdown, Metacritic score, playtime, and signals of social media coverage on Reddit,YouTube,Twitch. It also contains more novel and precise metrics such as the added_by_status column which complies how many users have the game marked as owned, playing, beaten, dropped, or to-play allowing us to assess player retention and interest over time.

Given the API raw-dump origin of this data, some issues arise. The dataset is extremely sparse: most entries (about 98%) correspond to low-information games with little or no rating, genre, or engagement data. We therefore filter the data to keep an information-rich subset without overwhelming noise. After restricting to games with a valid rating, genre, release date, and more than 10 ratings, we retain 12,207 games. Some columns are irrelevant for visualization (e.g. images), while others require preprocessing: missing values must be standardized, several fields contain JSON or semi-structured strings (`esrb_rating`, `added_by_status`, `ratings`), and multi-value fields such as genres, platforms, and tags are pipe-delimited, meaning that multiple values are stored in a single cell separated by delimiter characters like `|`.

These are standard cleaning steps, and the dataset remains high-quality and easy to filter for visualization purposes. The processing code creating a clean csv file can be found in rawg_data_cleaning.py

Problematic

Since the first game in 1958, the video game industry has structured itself and risen to become one of the top industries in the Entertainment sector, going through both hardships and glory. Among the countless games released over the past 40 years, only a handful have survived the passage of time. These special games are very few, yet they all share common characteristics that we, as video game enthusiasts and dedicated fans, want to highlight.

Our visualization therefore aims to reveal what makes these games special through dynamic and interactive visuals such as graph networks. The core idea is to display video games as nodes on a screen, enabling filtering and time navigation to observe how games rise and fall, and how some endure. From this general view, users could dive into more focused visualizations covering specific topics such as active player counts over time, and more.

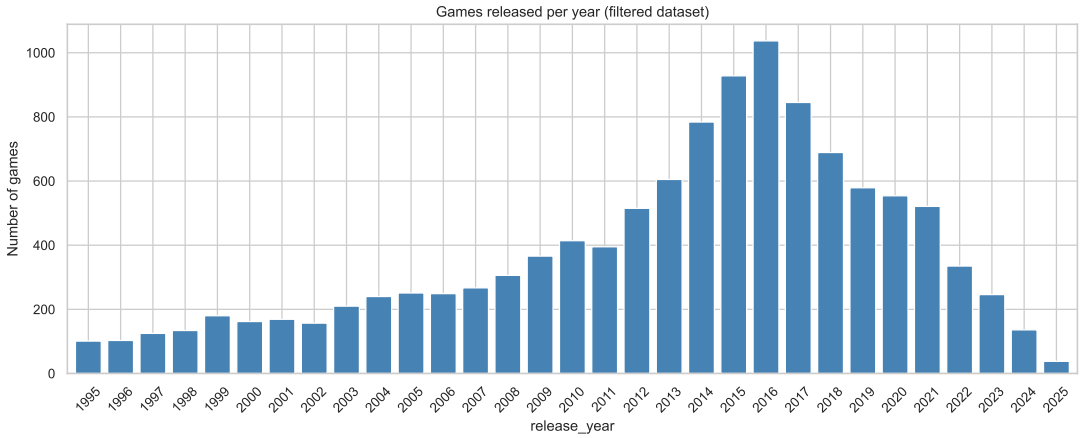
Our motivation stems from the fact that we are all still active players ourselves. As time reshaped the gaming landscape, we had the chance to play games that date back over a decade. We never fully understood why these games retained such attraction, why, despite outdated graphics and game mechanics that are no longer original, they still occupy a special place in the hearts of gamers.

Our target audience includes fellow video game enthusiasts, but also anyone curious about video games and their history. We place particular emphasis on accessibility, as video games are a universal medium for communication and shared enjoyment that transcends all boundaries.

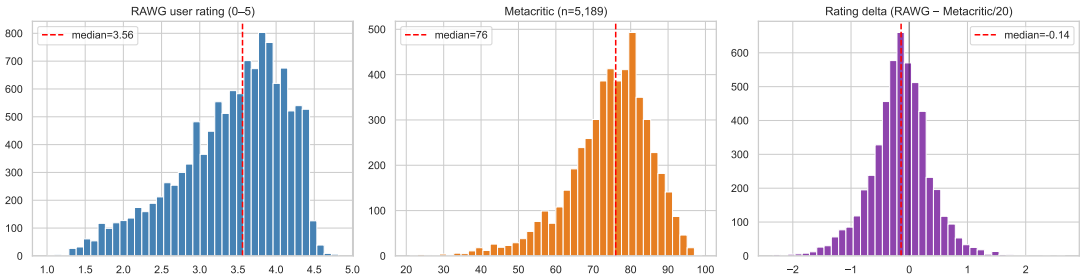
Exploratory Data Analysis

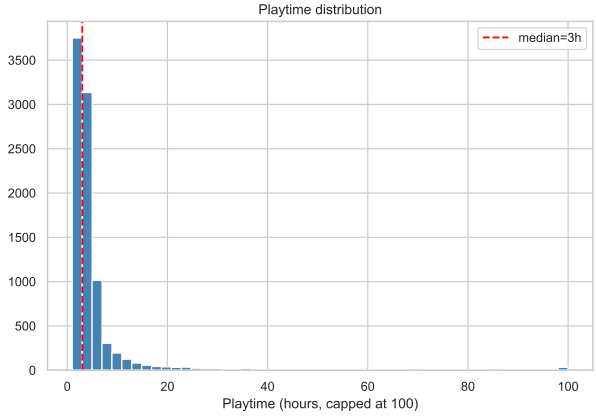
Using our cleaned csv file, we performed exploratory data analysis (full analysis at rawg_eda.ipynb) to identify both general trends and indicators of long-term game survivability:

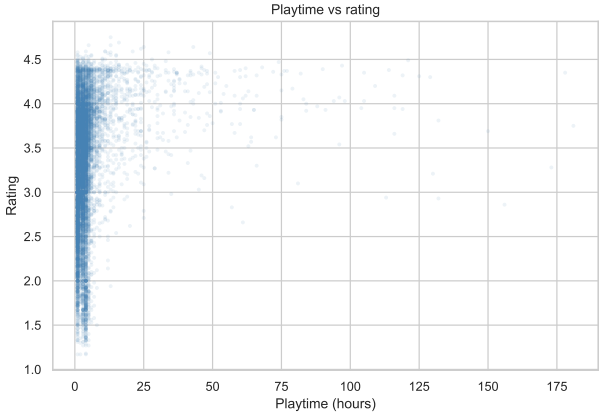
- Releases peak in 2016 (1,037 titles/year) and decline after 2020, likely because recent games have not yet crossed our ratings threshold (10 + reviews).



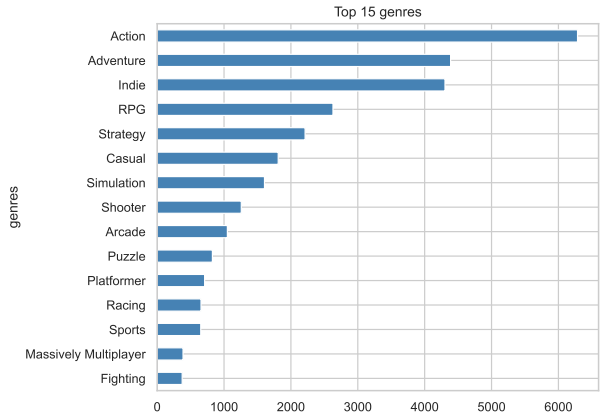
- RAWG scores skew positive (median 3.48/5), while Metacritic scores average 74/100 on the 42.5% of games where they are available. Playtime is highly skewed (median 3 hours) and shows a weak positive correlation with review scores.

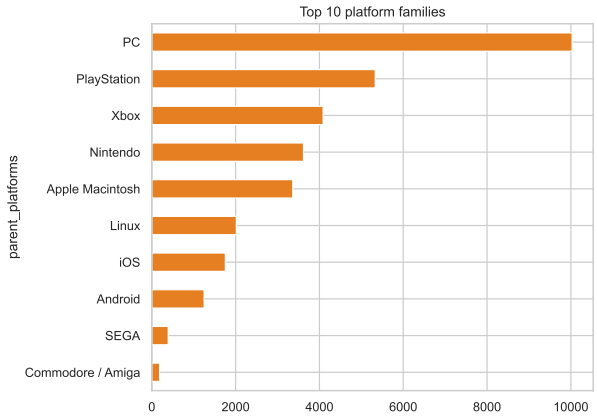




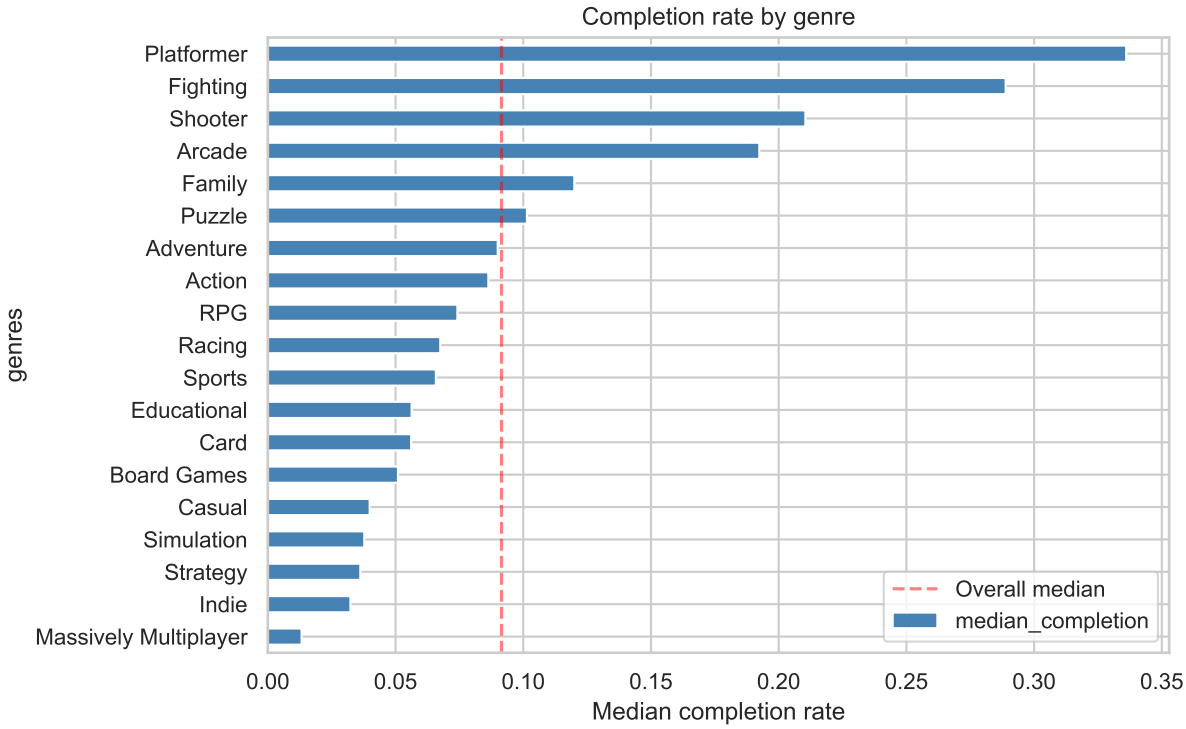


- Action is the most common genre (6,286 games), followed by Adventure and Indie. Games appear on 3 platforms and have 2.4 genres on average. Several variables remain sparse (50%+ null), so they are used only as supporting facets rather than primary axes of analysis.

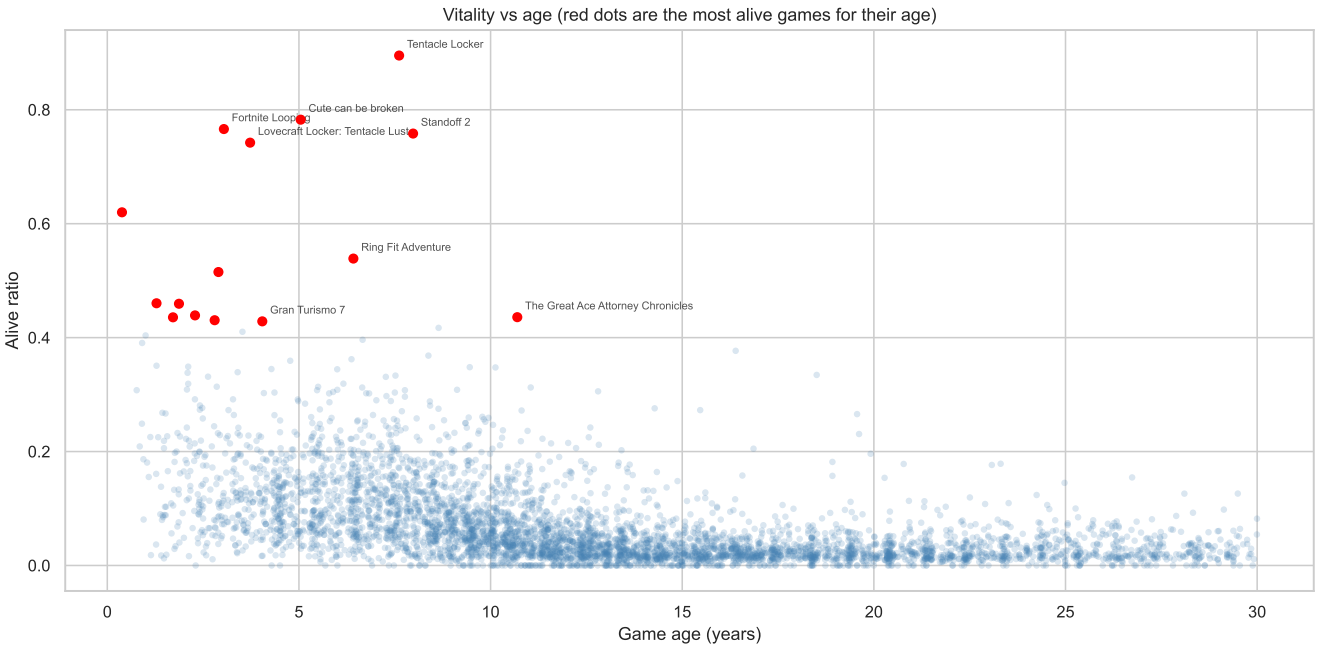




- The median completion rate, defined as `beaten/owned`, is 9.1%, while the drop rate, defined as `dropped/owned`, is 6.6%. Completion varies strongly by genre: narrative adventures and puzzles rank high, whereas MMOs, sports, and simulation games are rarely finished in the traditional sense

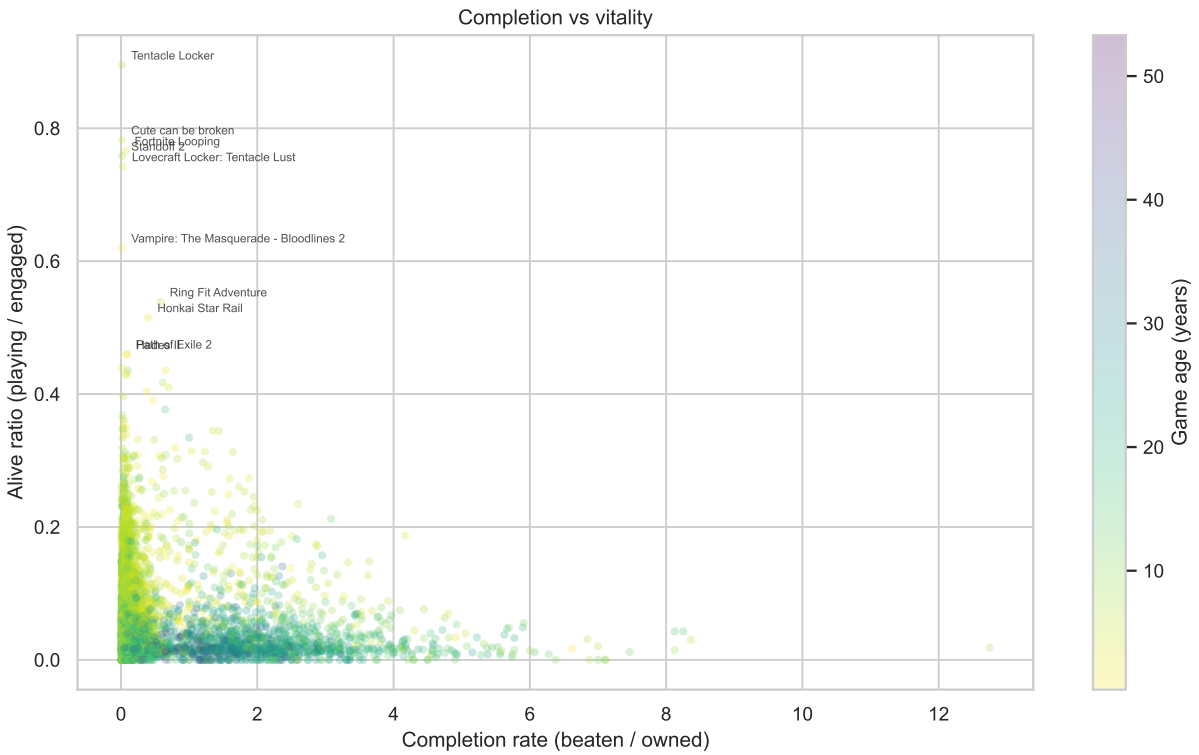


- We define `alive_ratio` (`playing/engaged`) as a measure of current vitality. Its median is 0.04, meaning only 4% of engaged players are still active for a typical game. Although vitality usually declines with age, some pre-2015 titles such as League of Legends, osu!, and Europa Universalis IV remain as active as recent releases.

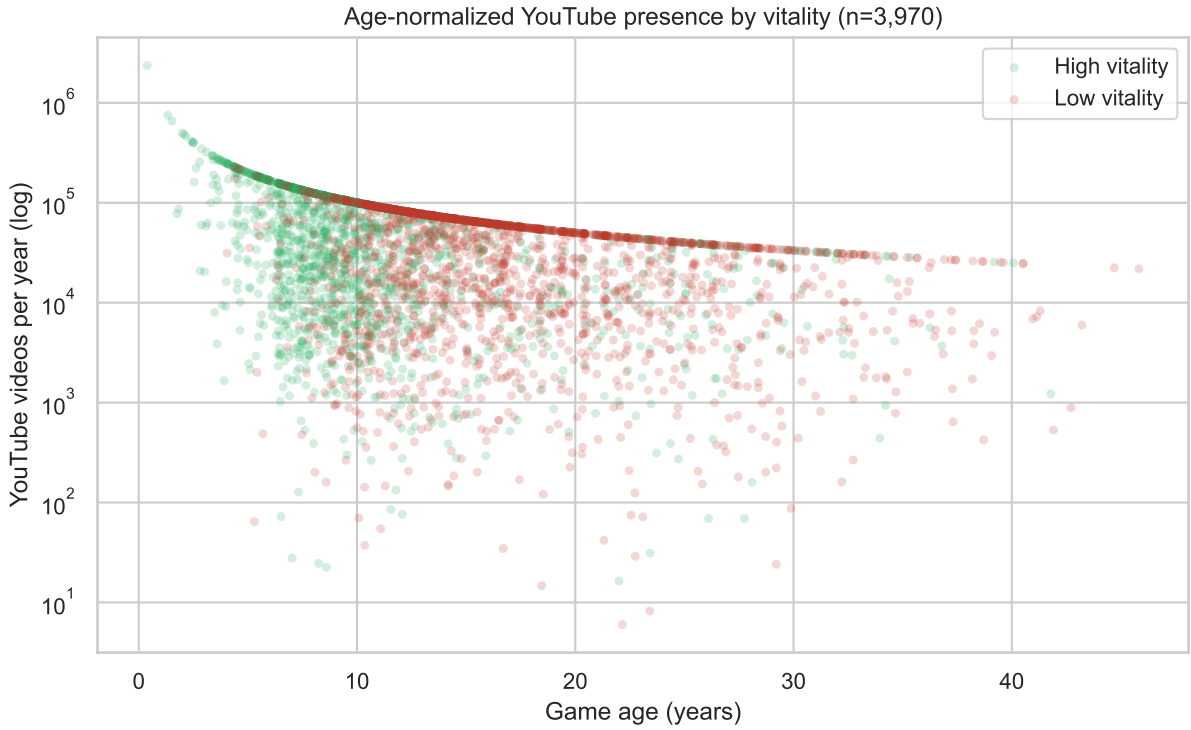


- Completion rate and `alive_ratio` are negatively correlated (-0.175): games designed not

to end, such as MMOs, simulation, and strategy titles, tend to retain active players longer than games built around completion.



- After normalizing YouTube activity by game age, games with high long-term vitality still receive about 1.5× more coverage per year than fading games.



These results confirm the feasibility of our project and provide a strong basis for an interactive visualization of game longevity which we'll aim to visualize in creative ways.

Related work

Existing RAWG dataset analyses on platforms like Kaggle and GitHub focus on static, descriptive statistics such as genre distributions and user ratings. For example, Kimchi21's analysis [4] explores genre preferences and engagement metrics through bar charts and summary tables. Similarly, an article by Victor Yamaykin [7] targets indie developers by analyzing completion and dropout rates by ESRB category. While valuable, these works predominantly focus on aggregate historical popularity and static analysis rather than isolating the indicators of long term player retention. Furthermore, they lack any form of interactive visualization. Meanwhile, the relevant academic literature [1] models playtime decay using Weibull distributions in telemetry data, aimed at researchers rather than a general audience.

Our approach differs from existing work primarily in terms of format and accessibility. By adopting a scrollytelling format, we translate game longevity into a narrative visualization, making it accessible to enthusiasts without requiring a statistical background. To the best of our knowledge, no existing project applies narrative visualization techniques to video game engagement data with this goal in mind.

In terms of visual and narrative direction, we draw conceptual and technical inspiration from three distinct paradigms. First, we adapt the temporal navigation from Histography [6] to map video games as nodes on a filterable timeline, allowing observation of parameter trends over time. Second, we consider the use of temporal force-directed graphs [2] to visualize how relationships between game clusters evolve over time. Finally, for the overall narrative, we draw inspiration from the scrollytelling format of The Pudding's *Is the Love Song Dying?* [5]. We intend to implement a similar packed bubble chart approach to transition the user from a general overview of the dataset down to the specific subset of games with high longevity.

References

[1] Christian Bauckhage, Kristian Kersting, Rafet Sifa, Christian Thurau, Anders Drachen, and Alessandro Canossa. How players lose interest in playing a game: An empirical study based on distributions of total playing times (pre-print). 09 2012.

[2] D3. Temporal force-directed graph. https://observablehq.com/@d3/temporal-force-directed-graph. Accessed: March 2026.

[3] Atalay Denknalbant. RAWG games dataset. https://huggingface.co/datasets/atalaydenknalbant/rawg-games-dataset, 2024. Accessed: March 2026.

[4] Kimchi21. RAWG.io Game Data Analysis. https://github.com/Kimchi21/RAWG.io_Game-Data-Analysis, 2023. Accessed: March 2026.

[5] David Mora and Michelle Jia. Is the love song dying? https://pudding.cool/2024/11/love-songs/, nov 2024. Accessed: March 2026.

[6] Matan Stauber. Histography. https://histography.io/, 2015. Accessed: March 2026.

[7] Victor Yamaykin. Indie games: a RAWG data analysis. https://medium.com/@victoryamaykin/indie-games-a-rawg-data-analysis-892b3b04e33e, 2023. Accessed: March 2026.